

CONSTRUCTION OS

PRD - Engineer Dashboard

Project-Centric Technical Engineering Role Workspace

Implementation-ready Product Requirements Document for Product, UX/UI, Frontend, Backend, Security, QA and DevOps Teams.

Product purpose: Define the exact Engineer / Technical Manager role shell, dashboard, assigned-project workspace, technical review surfaces, inspection integration and cross-module behavior. The experience is intentionally simple: open Dashboard, see assigned projects, see technical work requiring attention, enter a project and execute or review the work.

Field	Value
Document	PRD_Engineer_Dashboard
Version	1.0
Status	Implementation-ready role dashboard specification
Primary role	Engineer / Technical Manager
Base requirements	Construction OS Master PRD + PRD_2 Role Centers
Connected domains	Projects, Documents, Tasks, QA/QC Inspections, Approvals, Calendar/Meetings, BIM/3D, Work Progress
Platform alignment	Role Context, Role & Permission Matrix, Platform UI/UX, Approvals & Workflow
Scope	Engineer dashboard + exact sidebar + assigned-project technical workspace
Date	2026-08-07

Permanent dashboard scope lock

The Engineer Dashboard and Engineering role sidebar shall contain only the elements explicitly defined in this PRD. Developers must not add, infer, duplicate, substitute or auto-expose extra dashboard widgets, modules or shortcuts without an approved PRD revision. Platform Configuration may hide approved elements; it may not create unapproved ones.

Document Authority and Source Alignment

Priority	Authority	Controls
1	Complete Role & Permission Matrix	Authorization, scopes, deny-by-default behavior, field restrictions, revocation and audit.
2	Construction OS Master PRD - Architecture and Engineering Workspaces	Engineer responsibilities, assigned-project default, structural/civil/MEP scope, calculations, specifications, inspections, RFIs, submittals and coordination.
3	PRD_2 - Role Centers and Cues	Fixed role dashboard regions, Engineer primary cues, Work Inbox, operational content and recent activity.
4	Documents / Drawings / RFI / Submittal requirements	Controlled technical files, revisions, RFIs, submittals, Draft Mode and official current-file behavior.
5	QA/QC & Handover PRD	Inspection requests, inspection execution, measurements, evidence, result states and quality ownership.
6	Tasks / Calendar / Approvals / BIM / Work Progress PRDs	Execution, deadlines, decision orchestration, spatial coordination and project progress context.
7	Role Context Navigation PRD	Engineering shell remains active when shared modules are opened.
8	Platform UI/UX Architecture	Shared shell, progressive disclosure, responsive behavior, accessibility, motion and component patterns.
9	This PRD	Exact Engineer dashboard composition, sidebar and project-centric technical UX decisions.

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1. Executive Purpose

- Give Engineers one calm technical workspace centered on assigned projects and immediate engineering work.
- Reduce navigation depth: common engineering actions must be reachable within two navigation decisions.
- Show technical decisions and blockers before passive analytics: tasks, RFIs, inspections, approvals, calculations, submittals and coordination issues.

- Keep project and discipline context visible while moving between technical records, QA/QC, Documents, Tasks, Approvals, Calendar and BIM.
- Reuse authoritative source modules rather than creating a separate Engineering copy of project, inspection, file, task or approval records.

Final product position

Engineer opens Dashboard -> sees assigned projects -> sees technical work needing attention -> enters a project -> reviews, calculates, coordinates or inspects. Enterprise depth remains behind progressive disclosure.

2. Goals and Non-Goals

In scope	Out of scope
Exact Engineering sidebar and header	A separate Engineer application with copied modules
Fixed Engineer dashboard composition	Arbitrary dashboard drag-and-drop in v1
Assigned-project portfolio and project technical workspace	Company-wide project discovery without permission
Structural / Civil / MEP / Other discipline context	A new discipline database independent from Projects/Documents
Specifications, calculations, RFIs and submittals navigation	Replacing Documents/RFI/Submittal source records
Inspection projection and authorized inspection execution	Replacing QA/QC as quality-decision source of truth
Technical/BIM coordination	Treating every BIM clash as automatically accepted engineering work
Tasks, approvals, deadlines, meetings and activity projections	A second task, workflow or calendar engine inside Engineering

3. Core UX Principles

Principle	Required behavior
Project first	Assigned projects are the primary operating scope and the largest dashboard element.
Attention first	Actionable technical work appears before passive reporting.
Four primary cues	Use the four confirmed Engineer Role Center cues; secondary counts live inside work sections.
Discipline aware	Structural, Civil, MEP or Other filters narrow technical context without changing authorization.
Progressive disclosure	Dashboard is simple; registers and record pages contain full technical depth.
Context continuity	Selected project and discipline persist through Engineering pages until changed or unauthorized.
One source of truth	Dashboard/workspace displays projections and links; owning domains retain authoritative records.
No unrelated data	Engineering views do not expose unrelated Finance, HR, Legal or Sales fields through linked records.

4. Engineering Application Shell

Authenticated Application

- Active company context
- Active Engineering role/workspace context
- Active project + optional discipline context
- Engineering header + sidebar
- Dashboard / Engineering pages
- Permission-filtered shared source-module content

- Role/workspace determines the shell. Route determines page content. Permissions determine visible data and actions.
- Opening Projects, Documents, QA/QC Inspections, Tasks, Meetings, Approvals or BIM must never switch the user away from the Engineering shell.
- Browser refresh, direct links, notifications, global search and browser history preserve active Engineering context.
- A multi-role user remains in Engineering until explicitly switching active role.

4.1 Persistent top header

Control	Required behavior
Company	Active company identity; switching company revalidates role, project scope and sidebar.
Project Scope	All Assigned Projects or one authorized Project.
Discipline	All / Structural / Civil / MEP / Other configured engineering discipline; filter only, never an access grant.
Global Search	Permission-safe search across authorized technical and shared records.
+ New	Context-aware quick create exposing only authorized Engineering-relevant source actions.
Notifications	Permission-safe notifications deep-linked into the Engineering shell.
Profile / Role	Authenticated user and active role/workspace context.

4.2 Exact left sidebar

ENGINEERING

Dashboard
Projects

TECHNICAL

- Engineering Packages
- Specifications
- Calculations
- RFIs
- Submittals

SITE & COORDINATION

- Inspections
- Coordination
- Approvals

WORK

- Documents
- BIM / 3D
- Tasks
- Calendar

- This is the exact v1 Engineering sidebar hierarchy.
- Sidebar group labels are visual separators, not separate routes.

5. Dashboard Information Architecture - Exact Layout

Engineer Dashboard

- Header: All Assigned Projects | Project selector | Discipline | Search | + New
- Primary Cues: Technical Tasks | Open RFIs | Inspections | Approvals
- My Projects
- Needs My Attention
- Technical Status
- Upcoming
- Recent Activity

Layout lock

These seven dashboard regions are the complete v1 Engineer Dashboard. There is no generic company KPI section, financial panel, sales pipeline, arbitrary chart area or user-added widget region.

5.1 Full architecture map

ENGINEERING ROLE WORKSPACE

- Dashboard
 - Primary Cues
 - My Projects
 - Needs My Attention
 - Technical Status
 - Upcoming
 - Recent Activity
- Projects
 - [Assigned Project]
 - Overview
 - Engineering Packages
 - Specifications
 - Calculations
 - RFIs
 - Submittals
 - Inspections
 - Coordination
 - Approvals
 - Tasks
 - Documents
 - BIM / 3D
 - Activity
- Global Technical Registers
 - Engineering Packages
 - Specifications
 - Calculations
 - RFIs
 - Submittals
- Shared Operational Surfaces
 - QA/QC Inspections
 - Coordination / BIM Issues
 - Approvals
 - Documents
 - Tasks
 - Calendar
- Source Domains remain authoritative

6. Primary Cue Cards - Exact Set

Cue	Definition	Click-through
Technical Tasks	Authorized engineering Tasks requiring user action, including due/overdue technical assignments.	Tasks filtered to Engineering role scope and actionable state.
Open RFIs	Authorized open RFIs relevant to the Engineer in assigned-project scope.	RFI register with identical project/discipline/filter definition.
Inspections	Authorized inspection requests/inspections requiring Engineer action or attendance, based on QA/QC role/capability.	Shared Inspections view filtered to the same actionable set.
Approvals	Engineering-related review/approval work items currently assigned to the user.	My Approvals filtered to Engineering-related source records.

- Each cue shows current count, status meaning and click-through destination.
- Counts must reconcile exactly with the resulting filtered register.
- No additional top-level KPI cards are allowed in v1.

7. My Projects - Primary Dashboard Region

Project card element	Required content
Identity	Project name and optional project image/thumbnail from Projects when available.
Status	Canonical Project status and current technical/design stage when source context exists.
Discipline	Engineer-relevant discipline badge(s) where the assignment or project setup provides them.
Attention chips	Open RFIs, pending inspections, approvals, technical tasks or coordination issues - permission filtered.
Next deadline	Nearest authorized Engineering-related deadline from Tasks/Calendar/Workflow/inspection schedule.
Recent change	Latest meaningful Engineering activity for that project.
Action	Open Project Engineering Workspace.

No invented technical progress

Do not display a generic "engineering progress %" unless a governed metric exists in the authoritative project/work-progress source. Use canonical stage, status and actionable counts instead.

7.1 Project card behavior

- Default dashboard shows active assigned projects first.
- Cards remain compact; full project details belong in the Project Engineering Workspace.
- Discipline filter changes presentation/query scope but cannot reveal records outside authorization.
- If the Engineer has no assigned projects, show an empty state rather than company-wide projects.

8. Needs My Attention - Work Inbox

Type	Examples	Primary action
Technical Task	Calculation, review, design check, method/specification action due	Open Task / source record.

Type	Examples	Primary action
RFI	RFI assigned to Engineer or response due	Open RFI and respond where permitted.
Inspection	Inspection request assigned, scheduled, ready or needing result	Open QA/QC Inspection workspace.
Approval	Engineering review/approval assigned to user	Open approval workspace.
Submittal	Technical/material submittal requiring review or correction	Open source submittal.
Calculation	Calculation returned, review due or linked approval pending	Open controlled calculation record.
Coordination	BIM/technical coordination issue requiring action	Open issue/viewpoint or linked Task.
Clarification	Returned information request/comment requiring response	Open source record and respond.

- Order by governed priority, due date and risk; critical and overdue work cannot be buried by recent low-value updates.
- Each item identifies source module, project, record, current stage and due state.
- Work Inbox is a projection only; it never becomes a competing task, inspection or approval source of truth.
- Default visible views: My Priority, Due Today, Overdue and Waiting.

9. Technical Status

Status group	Dashboard summary
Engineering Packages	Open / In Preparation / In Review / Approved or current states supported by source records.
Specifications	Draft/current/review/returned counts based on controlled source status.
Calculations	Draft / Under Review / Returned / Approved-current where source state supports it.
RFIs	Open / Awaiting response / Overdue / Recently answered.
Submittals	Open / In Review / Approved / Rejected or Returned based on source state.
Inspections	Assigned / Scheduled / Ready / Result Issued / Reinspection Required when visible to the Engineer.
Coordination	Open engineering coordination issues, blocked items and waiting-on-others projections.

Status source rule

The dashboard must render source statuses. It must not invent an Engineering-only lifecycle that can disagree with Documents, QA/QC, Tasks, BIM or Workflow state.

10. Upcoming

Item	Source / behavior
Technical deadlines	Tasks / source record due dates.
Inspection schedule	Calendar projection from QA/QC inspection requests; QA/QC retains inspection status authority.
Review deadlines	Workflow work-item SLA/due information.
Meetings	Calendar/Meetings records relevant to assigned projects.

Item	Source / behavior
Scheduled submissions	Authorized submittal/specification/calculation deadlines represented by source records.

- Upcoming is a compact context panel, not a second Calendar or inspection scheduler.
- Clicking an item opens the authoritative Task, Meeting, Inspection, Approval or source record.

11. Recent Activity

- Engineering package/specification/calculation created, submitted, returned or approved.
- RFI assigned, answered or overdue.
- Submittal submitted, approved, rejected or returned.
- Inspection requested, scheduled, executed, failed, passed or reinspection required.
- Engineering approval decision or information request.
- BIM/coordination issue created, assigned, updated or resolved.
- Engineering-related task changed.
- Controlled technical document revision changed.

Activity is permission filtered

Activity items are projections of authoritative events. Revoked or unauthorized records disappear from the user-facing feed without deleting source audit history.

12. + New - Exact Quick Create Behavior

Quick create item	Rule
Engineering Package	Create only when the source domain supports package creation and user has project/discipline capability.
Specification	Create controlled source record/file flow in selected project.
Calculation	Create controlled technical calculation record in selected project/discipline context.
RFI	Create in selected/required project context.
Submittal	Only where Engineer is authorized to initiate the configured submittal type.
Inspection Request	Open QA/QC inspection-request flow only when user has request capability; does not create a local Engineering inspection.
Task	Open Tasks create flow with Engineering/project context prefilled.
Upload Document	Open Documents upload flow in authorized project/folder context.

- + New never bypasses required fields, approvals, Draft Mode, inspection readiness or source-module validation.
- When no project is selected, project-required create flows must ask for an authorized project.
- A BIM coordination issue is created through BIM/BCF context when applicable; it is not a generic Engineering quick-create unless the source flow explicitly supports it.

13. Projects Page - All Assigned Projects

Region	Required behavior
Scope	Only projects authorized for the current Engineer; assigned projects by default.
Default view	Visual cards optimized for fast project recognition and technical attention.
Alternate view	List view may be available using the shared register pattern.
Filters	Project status, discipline, technical stage, attention state and authorized company/branch context.
Card content	Project identity, status/stage, engineering attention items, next deadline and latest technical activity.
Open action	Enter the Project Engineering Workspace without leaving Engineering shell.

14. Single Project Engineering Workspace

Engineering > Projects > [Project]

Overview | Engineering Packages | Specifications | Calculations | RFIs | Submittals
Inspections | Coordination | Approvals | Tasks | Documents | BIM / 3D | Activity

Workspace header	Content
Project identity	Canonical project name, status and authorized project metadata.
Technical context	Active discipline context and configured technical stage when available.
Attention	Open engineering work, inspections, RFIs, approvals and blocked/returned items.
Next deadline	Nearest relevant technical/inspection/review deadline.
Quick actions	Context-aware actions from the approved + New set.

Project workspace rule

This page is an Engineering lens over one Project. It must not create duplicate project, file, task, QA/QC inspection, approval or BIM records.

14.1 Project Overview tab

- Project technical summary and current attention state.
- Engineering package/specification/calculation summaries.
- Open RFIs and submittals.
- Pending/ready inspections and reinspection requirements.
- Coordination issues and pending approvals.
- Upcoming deadlines/meetings and recent Engineering activity.

15. Engineering Packages

Element	Requirement
Identity	Package code/title, project, discipline and technical scope.
Contents	Links to controlled specifications, calculations, drawings/models, submittals or configured deliverables.
Owner / contributors	Authorized source ownership/assignment metadata.
Status	Source/project/workflow status; no separate unsupported lifecycle.
Review state	Embedded source/workflow projection where controlled review applies.
Due date	Source/task/workflow deadline when applicable.
Files	Open controlled technical files through Documents permissions.

Package ownership decision

Engineering Package is a project-centric technical grouping/work surface. It must reference authoritative source records and controlled Documents. It must not become a second file store or a second approval engine.

16. Specifications

Specification element	Requirement
Identity	Code/title, project, discipline, type/category and source record ID.
Scope	System/work package/location or project structure context where available.
Current revision	Current controlled revision and source status.
Author / reviewers	Authorized source metadata.
Approval state	Embedded source/workflow state only.
References	Linked drawings/models, calculations, RFIs, submittals, inspections or tasks where authorized.
Files	Controlled document access through Documents.

- Superseded specification revisions remain retrievable but must not be presented as current.
- Inspection readiness may reference exact current specification revision; stale references must be visible to the user.
- Engineering Dashboard does not duplicate specification content into local dashboard storage.

17. Calculations

Calculation element	Requirement
Identity	Calculation code/title, project, discipline and type.
Technical scope	System, structure, element, work package or location context where source structure supports it.
Inputs / assumptions	Controlled or referenced calculation assumptions and source data.
Author / checker	Authorized responsible engineer/checker metadata.
Revision / status	Current revision and source review status.
Review / approval	Workflow/source projection; current decision state only.
References	Linked specification, drawing/model object, RFI, submittal or task.

Calculation element	Requirement
Files	Calculation sheets/reports remain controlled Documents/source records.

Calculation governance

The dashboard may organize and surface calculation records, but it must not recalculate engineering design logic or silently overwrite authored technical results. Calculation validity remains with the controlled source record and professional review process.

18. RFIs

RFI element	Requirement
Context	Project plus linked drawing/specification/location/model object where applicable.
Question	Source RFI question.
Assignee	Authorized responsible user/role.
Due date	Source due date and derived overdue state.
Response	Authoritative response and response metadata.
Status	Source RFI status.
Links	Related technical record, task, document, approval, inspection or BIM object when authorized.

- Dashboard Open RFIs cue and Project RFIs tab reconcile to the same source query definition.
- Engineer may answer only RFIs for which the user has the required capability and scope.
- Overdue state is derived from due date/status; it is not a manually editable dashboard flag.

19. Submittals

- Show authorized material, technical-data, certification and engineering submittals relevant to the Engineer and selected discipline.
- Expose project, package/type, submitter, due/review state, approval status and linked controlled documents.
- Opening a submittal uses the shared source record and Approval Workspace; Engineering does not duplicate the record.
- Technical review may link to specifications, calculations, drawings, procurement/material context and inspections where authorized.

20. Inspections - QA/QC Integrated, Not Duplicated

Source-of-truth boundary

QA/QC owns formal inspection requests, inspection execution records, measurements, evidence and quality decisions. Engineering Dashboard exposes only permission-safe projections and approved actions. An Engineer may act as a configured inspector when authorized, but Engineering does not maintain a second inspection record.

Inspection element	Required Engineering view
Request identity	Request number, project, work package, contractor, discipline and inspection type.
Scope / location	Exact location, element/system and quantity where provided.
References	Current drawing/specification/method statement and exact controlled revisions.
Schedule	Requested/planned date/time and Calendar projection.
Readiness	Prerequisites, approved documents, materials/tests, access, evidence and required attendees.
Assignment	Requester, assigned inspector and required participants.
Status	QA/QC lifecycle status including Ready, In Inspection, Result Issued, Reinspection Required and alternatives.
Result	Pass, Pass with Comments, Conditional Acceptance, Partial Pass, Fail or other source result.
Evidence	Checklist, measurements, photos/files, witnesses, findings and linked corrective records.

20.1 Engineer inspection behavior

- When configured as inspector, Engineer may open the shared Inspection workspace and record permitted checklist, measurements, evidence and result through QA/QC APIs.
- A failed/rework-required result follows QA/QC/Task corrective workflow and preserves prior inspection history.
- A result cannot be rewritten from Fail to Pass without reinspection or an authorized superseding process.
- Scheduling integrates with Calendar, but QA/QC remains authoritative for inspection status.
- The Dashboard Inspections cue must reconcile with the exact actionable inspection list.

21. Coordination

Coordination source	Engineering behavior
BIM / BCF issue	Open coordination issue, saved viewpoint, model context, discipline, priority and status from BIM source.
Task	Executable assignment remains owned by Tasks; Engineering shows status projection and deep link.
RFI	Clarification remains owned by RFI/source document workflow.
QA/QC issue	Formal NCR/defect remains owned by QA/QC; Engineering may link and respond where authorized.
Meeting / decision	Meeting/decision source remains authoritative; Engineering shows linked context.

- BIM owns coordination issue/viewpoint and BCF exchange records; Tasks owns executable assignments created from issues.
- A clash import is not automatically an accepted engineering issue. Acceptance/triage follows BIM/coordination policy.
- Coordination page may group issues by project, discipline, location, status, responsible party and due date without copying source records.

22. Approvals

Approval element	Requirement
Source record	Exact engineering record/version under review.
Stage	Current workflow stage and remaining decision path.
Participants	Resolved approver/reviewer participants within permitted visibility.
Due / SLA	Workflow-owned timer and escalation state.
Decision	Approve/reject/return/request information only through Workflow capability.
Version change	Material changes invalidate/restart configured stages as Workflow determines.
Finalization	Approval is not complete until source module confirms resulting transition.

Approval engine boundary

Source modules own what is being decided and the business consequence. Approvals & Workflow owns routing, participants, decisions, escalation and audit. Engineer Dashboard displays and opens this workflow; it does not implement a private approval chain.

23. Documents

- Documents remains the controlled source for technical files, revisions, comments, comparison, approvals and official issue state.
- Engineering views may filter technical folders, project documents, specifications, calculation reports, drawings and submittal files without copying them.
- Draft files follow controlled Draft Mode and selected-collaborator visibility where source policy applies.
- Superseded revisions remain preserved and clearly distinguished from current official versions.
- Project Tranzit may be used as a collaborative temporary workspace where authorized; promotion to official folders preserves identity/history.

24. BIM / 3D

Capability	Engineering behavior
Navigate	Project -> building/block -> level -> zone -> unit/room/system -> object.
Select object	Show authorized drawings, specifications, RFIs, submittals, tasks, inspections and issues.
Viewpoint / markup	Open or create governed viewpoint/markup where BIM capability allows.
Coordination issue	Create/link BIM issue; executable work is linked to Tasks, not copied.
Model revision	Stable object links survive revisions where BIM identity rules confirm continuity.

BIM rule

BIM owns visual/spatial truth of model versions and object relationships. Operational modules own business facts. The Engineer Dashboard treats BIM as a navigation and coordination surface, never as an alternative source of record.

25. Tasks and Calendar

25.1 Tasks

- Tasks is the common execution layer for engineering assignments, corrections, coordination work and follow-up actions.
- Needs My Attention and Technical Tasks cue query Tasks/workflow/source states; they do not copy task status locally.
- Task privacy, participants, approvals, evidence and timeline remain owned by Tasks.
- A failed inspection may create/link corrective Task while QA/QC retains the inspection decision.

25.2 Calendar and Meetings

- Calendar owns personal agenda, availability and calendar projections.
- Meetings owns meeting records, agenda, attendance, minutes and decisions.
- Inspection scheduling may appear in Calendar, but QA/QC owns inspection status.
- Upcoming shows only compact authorized projections and deep links.

26. Data Ownership and Source-of-Truth Boundaries

Domain	Owns	Engineering Dashboard role
Projects	Project identity, structure, membership/assignment and project context	Filter, summarize and deep-link.
Documents	Technical files, revisions, official versions, controlled document metadata	Display/open controlled records; no file duplication.
RFIs / Submittals	Questions/responses and technical submission state	Display, filter and act through source APIs.
QA/QC	Inspection requests, inspections, evidence, quality decisions, NCR/defect records	Project/action projection; authorized inspector uses QA/QC workspace.
Tasks	Assignments, status, dependencies, evidence and execution timeline	Work Inbox / task projection.
Approvals & Workflow	Routing, work items, decisions, delegation, SLA and escalation	Decision projection / approval workspace.
BIM / 3D	Models, model versions, objects, viewpoints, markups and BIM coordination issues	Spatial navigation and coordination.
Calendar / Meetings	Agenda/time projection, meetings, minutes and decisions	Upcoming / meeting links.
Work Progress	Executed quantities, progress, schedule/constraint records	Read technical/project progress context only where authorized.

Permanent integration rule

Engineering may display data from connected modules, but it must not create editable local copies of another module's authoritative records.

27. Authorization and Permission Behavior

- Engineer sees assigned/authorized Projects by default. Same-company membership alone does not grant project access.
- Every sidebar item, dashboard count, card, row, field, activity event and deep link is permission filtered by backend authorization.
- Discipline filter never expands permissions; it only narrows authorized data.
- Inspection execution appears only when the user has the applicable QA/QC/inspection capability and project scope.
- Approvals appear only for decision work items the user is permitted to know about and act on.
- Restricted related records may appear only as safe locked relations where platform policy explicitly permits this; sensitive content remains absent.

27.1 Personal dashboard isolation

Dashboard context key
tenant_id + company_id + user_id + active_role_id + project_scope + authorization_version

- Backend derives user identity from the authenticated session; never trust client-supplied user ID for dashboard ownership.
- Cache/read-model keys include tenant, company, user/effective authorization, active role and project/discipline filters where applicable.
- Late responses from a previous company/project/role context must not overwrite the current context.
- Access revocation clears stale client state and invalidates relevant cached projections.

28. Shared-Module Navigation and Role Context

Navigation path	Required result
Projects	Project opens inside Engineering shell; Engineering sidebar/header remain active.
Documents	Technical document opens inside Engineering shell; source page remains reusable.
Inspection	QA/QC inspection page opens inside Engineering shell where route architecture supports shared content.
Task	Task opens inside Engineering shell.
Approval	Approval workspace opens inside Engineering shell.
Meeting	Meeting opens inside Engineering shell.
BIM	BIM viewer opens in Engineering context with permissions unchanged.

Role-context rule

A module must never decide the role dashboard. The active authenticated role/workspace determines the shell; the route determines content.

29. Platform Configuration

- All approved dashboard/sidebar elements remain compatible with Platform Configuration at module, page, section, widget, feature, action and report level.
- Configuration may hide or disable approved items and configured disciplines/subtypes; it may not create new unapproved dashboard elements.
- Status labels and workflows remain governed by their source modules/configuration.
- Configured feature removal must reflow layout cleanly without empty cards, dead routes or broken counts.

30. Frontend Architecture

```

EngineeringShell
├── RoleHeader
├── EngineeringSidebar
├── RouteOutlet
│   ├── EngineerDashboard
│   ├── AssignedProjectsRegister
│   ├── ProjectEngineeringWorkspace
│   └── SharedSourceModulePages
└── ContextStores
    ├── Auth / active role / company
    ├── Project + discipline scope
    ├── Permission capabilities
    └── Query/cache state
  
```

30.1 State and routing requirements

- Role shell is mounted independently from shared-module route content.
- Project and discipline selection are explicit URL/session context where appropriate and survive normal navigation.
- Shared route pages must not import or force another role layout.
- Queries include context versioning/cancellation so stale responses cannot replace newer scope.
- Dashboard widgets use common query/read-model primitives and shared status components.

31. Logical API / Read-Model Contract

Endpoint / capability	Purpose
GET /engineer/dashboard	Return permission-filtered cue counts, My Projects, attention, technical status, upcoming and activity.
GET /engineer/projects	Return authorized assigned-project portfolio with technical attention summaries.
GET /engineer/projects/{projectId}/overview	Return project Engineering summary projection.
GET /engineer/projects/{projectId}/technical-status	Return source-linked package/spec/calculation/RFI/submittal/inspection/coordination summaries.
GET /engineer/inbox	Return Engineering Work Inbox projection with source identifiers and actions.
GET /engineer/quick-create	Return authorized quick-create options for current context.

- These are logical contracts; final API registry may choose different route names while preserving behavior.
- Every response includes context/freshness metadata sufficient to reject stale role/project results.
- Dashboard read models store projections only and never become editable business truth.
- Drill-through counts must use the same governed filters as destination lists.

32. Responsive UX and Motion

Viewport	Required behavior
Desktop	Persistent/collapsible sidebar; project cards, attention queue and technical status use responsive grid.
Tablet	Sidebar becomes compact/drawer; project/discipline context remains visible; tables adapt or use horizontal-safe register pattern.
Mobile	Work Inbox/Needs My Attention appears before secondary status; navigation becomes drawer/bottom-safe pattern; actions remain reachable.

- Use ConstructionOS premium light design system, restrained gold accent and dark-grey typography.

- Use 120-160 ms micro-interactions, 180-220 ms drawers and 200-250 ms page transitions; no bounce/parallax/long fades.
- Honor reduced-motion preferences.
- Core flows target WCAG 2.1 AA; status meaning never depends on color alone.

33. Empty, Loading, Error and Partial Failure States

State	Requirement
No assigned projects	Explain that no projects are currently assigned/authorized; do not show all company projects.
No attention items	Positive empty state: no current technical action required.
No packages/specs/etc.	Contextual empty state with create action only when authorized.
No inspections	Show no actionable/relevant inspections; do not infer all are passed.
Loading	Skeletons match final geometry; preserve shell and selected project/discipline.
Partial failure	Healthy regions remain usable; failed region identifies retry without fabricating zero values.
Access revoked	Remove inaccessible content immediately, clear stale state and show permission-safe result.
Source unavailable	Show source-unavailable state; never use stale data as current without freshness indicator.

34. Performance, Freshness, Audit and Notifications

34.1 Performance and caching

- Typical authenticated dashboard initial load target: ≤ 2.5 seconds on supported broadband under normal operating conditions.
- Key dashboard API reads target p95 ≤ 500 ms excluding heavy files/models/reports.
- Use project/role-scoped read models for dashboard summaries; do not calculate all counts from raw records on the client.
- Cache keys include tenant, company, user/effective authorization, active role and project/discipline scope as applicable.
- BIM model geometry streaming is not a blocking dependency for initial dashboard load.

34.2 Audit, security and observability

- All mutations occur through source modules and retain their source audit requirements.
- Workflow decisions are immutable and source finalization is traceable.
- Inspection result history remains QA/QC-authoritative and cannot be silently rewritten by Engineering.
- Dashboard must not expose restricted content through counts, tooltips, search, notifications or activity text.
- Log authorization failures, stale-context suppression, projection refresh failures and source-module partial failures using tenant-safe diagnostics.
- No cross-tenant, cross-company or cross-user dashboard data leakage is acceptable.

34.3 Notification behavior

- Notifications deep-link to permitted source records inside the active Engineering shell.
- Engineering-relevant examples include RFI assignment/answer/overdue, inspection ready/result/reinspection, submittal decision, approval request, technical task change and coordination issue update.
- Notification delivery never grants record access and must stop when access is revoked.
- Duplicate event noise is grouped/de-duplicated according to Notification/Event Centre policy.

35. Testing Strategy

35.1 Unit / component tests

- Exact four cue definitions and click-through filters.
- Project card and Work Inbox sorting/empty states.
- Discipline filtering without permission expansion.
- Technical Status mapping to source statuses.
- Permission/configuration-hidden reflow.
- Quick Create option resolution.

35.2 Integration tests

- Projects assignment and project-scope filtering.
- Documents current revision/supersession behavior for technical files.
- RFI/submittal source-link reconciliation.
- QA/QC inspection request/list/detail/action projection and Calendar schedule integration.
- Failed/reinspection flow linking to Tasks without duplicated inspection state.
- Tasks Work Inbox projection.
- Approvals work item and source-finalization state.
- BIM coordination issue and task relationship behavior.
- Calendar/meeting Upcoming projection.

35.3 Security / context tests

- Engineer never sees unassigned project data without explicit access.
- Engineering shell persists on shared Project, Task, Document, Inspection, Meeting, Approval and BIM routes.
- Direct URLs, refresh, notifications, recent items and global search preserve Engineering role context.
- Role switch changes shell explicitly; opening a module never changes role.
- Cross-user, cross-company and cross-tenant negative isolation tests pass.
- Unauthorized Finance/HR/Legal/Sales fields are absent from API responses.
- Discipline filter cannot reveal unauthorized project/record data.

36. Acceptance Criteria

ID	Acceptance criterion
ED-AC-01	Engineer login opens the fixed Engineering Dashboard and Engineering shell.
ED-AC-02	Dashboard contains exactly Primary Cues, My Projects, Needs My Attention, Technical Status, Upcoming and Recent Activity.
ED-AC-03	Primary cues are exactly Technical Tasks, Open RFIs, Inspections and Approvals.
ED-AC-04	Only authorized/assigned projects appear by default.
ED-AC-05	Project card opens the Project Engineering Workspace without leaving Engineering shell.
ED-AC-06	Project workspace tabs match this PRD and preserve project/discipline context.
ED-AC-07	Needs My Attention reconciles with Tasks, QA/QC, Workflow and source records and never stores duplicate work state.
ED-AC-08	Engineering package/specification/calculation status uses source-controlled records and revisions.

ID	Acceptance criterion
ED-AC-09	RFI counts and project RFI lists reconcile.
ED-AC-10	Submittal status uses authoritative source/workflow state.
ED-AC-11	Inspections are QA/QC-owned; Engineering projection and authorized inspector actions reconcile with QA/QC.
ED-AC-12	Failed inspection cannot be shown as passed without source-authorized reinspection/supersession.
ED-AC-13	Coordination page preserves BIM issue ownership and Task execution ownership.
ED-AC-14	Approvals use the platform Workflow engine and show current source finalization state.
ED-AC-15	Documents use shared controlled storage with no duplicate file copy.
ED-AC-16	BIM is navigation/coordination only and opens linked authoritative records.
ED-AC-17	Tasks and Calendar are shared source modules; dashboard shows projections only.
ED-AC-18	Opening any shared module preserves Engineering header/sidebar/active role.
ED-AC-19	Permissions are enforced on backend responses, fields, counts, deep links and activity.
ED-AC-20	Discipline filters never expand access.
ED-AC-21	Disabled features reflow cleanly with no dead links or blank spaces.
ED-AC-22	Desktop, tablet and mobile layouts remain usable and WCAG 2.1 AA compliant.
ED-AC-23	No dashboard data crosses users, companies, tenants or stale authorization contexts.
ED-AC-24	No unapproved dashboard widget or sidebar item is present.

37. Explicitly Not Allowed

- Additional Engineer dashboard KPI cards, charts or widgets not defined in this PRD.
- A generic company KPI dashboard inside Engineering.
- Finance, payroll, legal or sales pipeline widgets on the Engineer Dashboard.
- Showing all company projects to an Engineer solely because they are in the same company.
- Duplicating Projects, Documents, RFIs/Submittals, QA/QC, Tasks, Workflow, Calendar, Work Progress or BIM business records in Engineering storage.
- Creating an Engineering-only inspection state that can disagree with QA/QC.
- Automatically converting every BIM clash into an accepted engineering issue/task.
- Hard-coded role checks that replace the platform capability/scope authorization model.
- A shared module changing the user into another role shell.
- User rearrangement of dashboard regions in v1.
- Generic engineering-progress percentages without a governed metric definition.
- Treating BIM model properties as authoritative operational business state.

38. Definition of Done

1. Exact Engineering sidebar, header and Dashboard layout are implemented.
2. Assigned-project portfolio and Project Engineering Workspace are fully connected.

3. Engineering packages, specifications, calculations, RFIs, submittals, inspections, coordination, approvals, Documents, BIM, Tasks and Calendar deep links resolve to authoritative source records.
4. QA/QC inspection ownership and authorized Engineer inspection behavior pass reconciliation and history tests.
5. Role/context persistence and permission filtering pass all shared-route and direct-link tests.
6. Dashboard scope lock is respected with zero unapproved elements.
7. Responsive, accessibility, loading, empty, error and partial-failure states pass visual QA.
8. Performance and cache isolation targets pass with representative enterprise data.
9. Security, audit and negative isolation tests pass with zero critical defects.

Final implementation rule

If an element is not defined in PRD_Engineer_Dashboard, it does not appear on the Engineer Dashboard or Engineering sidebar. New dashboard content requires an approved PRD revision.

Appendix A. Source Alignment Matrix

PRD decision	Source alignment
Assigned projects first	Master PRD Engineering workspace rule: Engineers see relevant modules and assigned projects by default.
Four primary cues	PRD_2 Engineer Role Center: Technical Tasks, Open RFIs, Inspections and Approvals.
Engineering packages / calculations / coordination / deadlines	PRD_2 primary operational content for Engineer Role Center.
Structural / Civil / MEP / calculations / specifications / inspections	Master PRD Architecture and Engineering Workspaces.
Controlled technical files, RFIs and submittals	Master PRD Documents, Drawings, Revisions, RFIs and Submittals requirements.
Inspection ownership / lifecycle / measurements / evidence	QA/QC & Handover PRD.
Engineering or QA/QC inspector can receive ready-for-inspection work	Task Orchestration inspection workflow.
Work Inbox	Platform UI/UX Role Center and Work Inbox architecture.
Approvals projection	Approvals & Workflow Engine separation of business record and decision orchestration.
Tasks	Tasks Module as common execution layer with linked source context.
Upcoming / Meetings	Calendar owns agenda projections; Meetings owns meeting records.
BIM / Coordination	BIM owns spatial/model truth and coordination issues; Tasks owns executable assignments.
Engineering shell persistence	Role Context and Shared Module Navigation PRD.
Fixed layout / no drag-drop	Master dashboard rules and Platform UI/UX baseline.

Appendix B. Role-to-Source Architecture Map

Engineer Dashboard / Engineering Shell

```

| Projects ----- owns project context / assignment
| Documents ----- owns controlled technical files / revisions
| RFI & Submittal source - owns questions / submissions / statuses
| QA/QC ----- owns inspections / evidence / quality decisions
| Tasks ----- owns execution / responsibility / corrective work
| Workflow ----- owns approval routing / decisions / SLA
| BIM ----- owns model versions / objects / coordination issues
| Calendar / Meetings ---- owns time projections / meeting records
| Work Progress ----- owns executed progress / measurements / constraints

```

Engineering Dashboard = permission-filtered read model + deep links + source actions
 Engineering Dashboard != new source of business truth

Appendix C. Route and Context Map

Example route	Engineering context behavior
/app/engineering	Engineer Dashboard in Engineering shell.
/app/projects	Projects register filtered by Engineering authorization/assignment while Engineering shell remains active.
/app/projects/{projectId}	Project page/workspace opened in Engineering shell with project context.
/app/rfis/{id}	Shared RFI record opened in Engineering shell.
/app/inspections/{id}	Shared QA/QC inspection record opened in Engineering shell where authorized.

Example route	Engineering context behavior
/app/tasks/{id}	Shared Task record opened in Engineering shell.
/app/approvals/{id}	Approval workspace opened in Engineering shell.
/app/bim/{projectId}	BIM viewer opened in Engineering shell with project/model permissions.

Route implementation principle

The exact URL scheme may vary in the API/route registry. The invariant is mandatory: active Engineering role/workspace resolves the shell; the route resolves shared content; permissions resolve data and actions.